7.2 Stretching Without Turning

WORKSHEET

The question

STEP III 1990, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q6

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself What does it mean for T to leave a vector "unchanged in direction but multiplied by a scalar"? Write it as an equation.

STAGE 1 The eigenvectors

Covers: the two vectors left unchanged in direction

Ask yourself Apply the matrix to $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and to $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$. Is each answer a multiple of the vector you started with? Which multiple?

STAGE 2 The image of the circle

Covers: the equation of E, and its area

Ask yourself Invert the matrix, so that x and y are written in terms of X and Y. Substitute into $x^2 + y^2 = 1$ and tidy up.

Pause and try For the area: a linear map multiplies every area by the absolute value of its determinant. What is the area inside C? You can check your answer using the two stretch factors from the first stage.

STAGE 3 The greatest value of X

Covers: the greatest value of X on E

Ask yourself Write X in terms of the point (x, y) on C. How large can $9x - 2y$ be when $x^2 + y^2 = 1$? Think of it as a dot product with a unit vector.

STAGE 4 A tangent

Covers: the tangent to E

Pause and try The tangent to C at $\frac{1}{5}(3,4)$ is $3x+4y=5$. A linear map sends tangent lines to tangent lines, so substitute your expressions for x and y in terms of X and Y into this line.

What's really going on

The spectral theorem

Every real symmetric matrix has a full set of perpendicular eigenvectors, with real eigenvalues. This is the *spectral theorem*, one of the central results of a first linear algebra course. It says that every symmetric linear map is, in suitable perpendicular coordinates, just a stretch along each axis. Equivalently, every quadratic form like $8X^2+12XY+17Y^2$ can be rotated into the form $\lambda_1 u^2+\lambda_2 v^2$, which is why every central conic is an ellipse or hyperbola with perpendicular axes (the *principal axis theorem*). The idea goes back to Cauchy in the 1820s, who used it to find the principal axes of quadric surfaces and of the stress in a solid body.

Symmetry is what makes this work. A matrix that is not symmetric need not be so obliging: a rotation through 90° turns every direction, so it has no real eigenvectors at all, and the shear $\begin{pmatrix}1 & 1\\ 0 & 1\end{pmatrix}$ has only one eigenvector direction, so it is not a stretch along any pair of axes.

Eigenvectors everywhere

Here are a few of the places eigenvectors appear.

- **Vibrations.** A system of masses and springs vibrates in *normal modes*, patterns in which every part moves in step at a single frequency. These are eigenvectors of a matrix, and the frequencies come from the eigenvalues (Section 11.2).
- **Data.** The directions in which a cloud of data points is most spread out are the eigenvectors of its covariance matrix. Finding them is *principal component analysis*,

one of the basic tools of statistics and machine learning (Section 13.7).

- **Quantum mechanics.** The possible results of measuring a quantity are the eigenvalues of a matrix (or operator), and the energy levels of an atom are eigenvalues.
- **Web search.** Google's PageRank, the original idea behind the search engine, ranks pages by the entries of the eigenvector, with eigenvalue 1, of a huge matrix describing the links between pages.

Try this next

1. **Finding eigenvalues.** Show that the eigenvalues of $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ are the roots of $\lambda^2 - (a+d)\lambda + (ad - bc) = 0$, and check that the matrix in the question gives $\lambda = 2$ and 4.
2. **Perpendicular eigenvectors.** Show that if $\mathbf{M}$ is symmetric and $\mathbf{Mu} = \lambda\mathbf{u}$, $\mathbf{Mv} = \mu\mathbf{v}$ with $\lambda \neq \mu$, then $\mathbf{u} \cdot \mathbf{v} = 0$. (*Hint:* compute $\mathbf{v} \cdot \mathbf{Mu}$ in two ways.)
3. **A rotation.** Show that the rotation matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$ has no real eigenvectors unless θ is a multiple of π. What are its complex eigenvalues?
4. **Powers.** Using the eigenvectors, find $T^n \begin{pmatrix} 1 \\ 0 \end{pmatrix}$. (*Hint:* write $(1, 0)$ as a combination of $(1, 2)$ and $(2, -1)$.)

Further reading

- **Gilbert Strang, *Introduction to Linear Algebra*** (Wellesley-Cambridge). Chapter 6 is an excellent introduction to eigenvalues and their applications.
- **Kurt Bryan & Tanya Leise, "The $25,000,000,000 eigenvector: the linear algebra behind Google"**, *SIAM Review* 48 (2006). Readable and freely available.
- **3Blue1Brown, *Essence of Linear Algebra*** (video series, free online). Beautiful animations of exactly the geometric ideas in this section.

The full guided solution is in Section 7.2:
https://beyondstep.keithmcnulty.org/ch07/eigenvectors.html